Model Question Paper – Cloud Data Engineer - Jain University

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SQL File 3*

Limit to 1000 rows

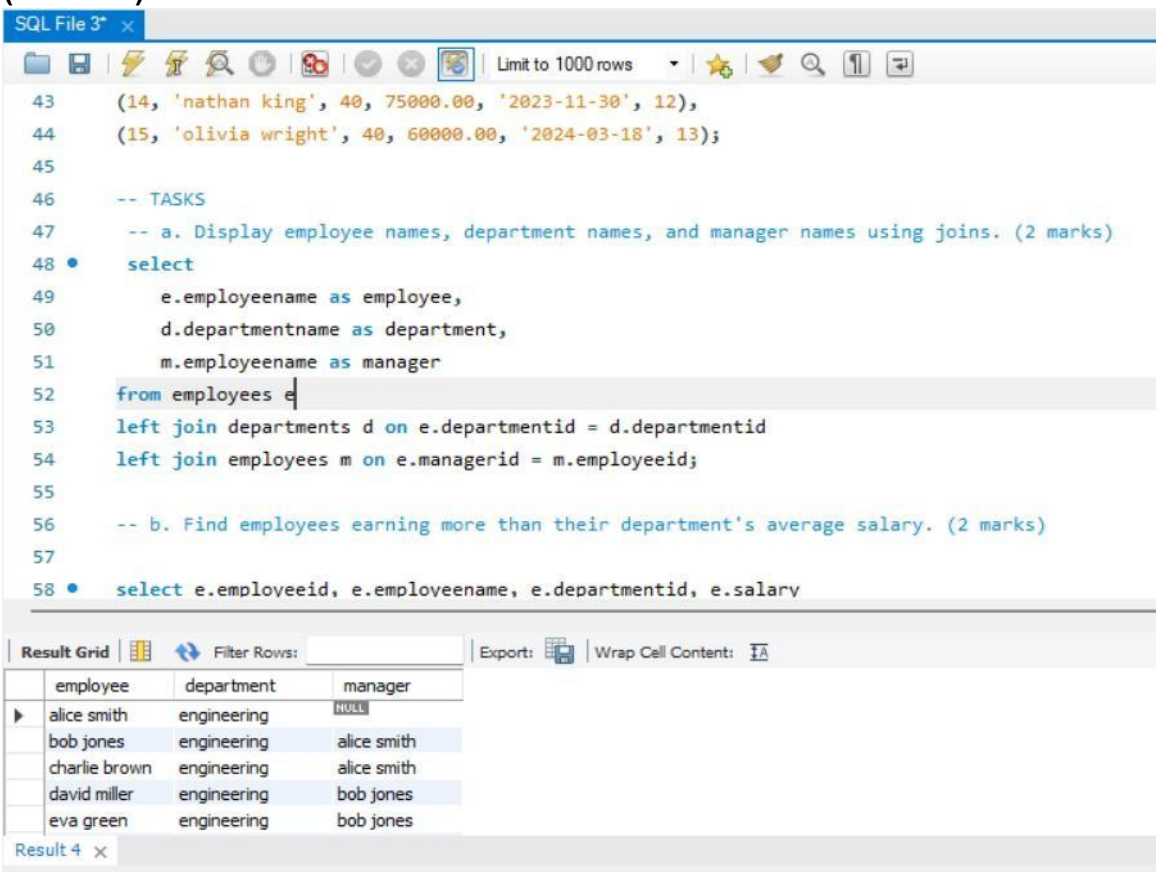
```sql
create database EnterpriseDB;
use EnterpriseDB;

create table Departments (
    DepartmentID int Primary Key,
    DepartmentName varchar(100) not null,
    Location varchar(100) not null
);

create table Employees(
                EmployeeID int Primary Key,
                EmployeeName varchar(100) not null,
                DepartmentID int,
                Salary decimal(10,2) not null ,
                JoiningDate DATE not null,
                ManagerID int,
                 foreign key (DepartmentID) references Departments(DepartmentID),
                  foreign key  (ManagerID) references Employees(EmployeeID)
                );


insert into departments (departmentid, departmentname, location) values
(10, 'human resources', 'new york'),
```

a. Display employee names, department names, and manager names using joins. **(2 marks)**



```
SQL File 3*
Limit to 1000 rows
43      (14, 'nathan king', 40, 75000.00, '2023-11-30', 12),
44      (15, 'olivia wright', 40, 60000.00, '2024-03-18', 13);
45
46      -- TASKS
47       -- a. Display employee names, department names, and manager names using joins. (2 marks)
48      select
49          e.employeename as employee,
50          d.departmentname as department,
51          m.employeename as manager
52      from employees e
53      left join departments d on e.departmentid = d.departmentid
54      left join employees m on e.managerid = m.employeeid;
55
56      -- b. Find employees earning more than their department's average salary. (2 marks)
57
58      select e.employeeid, e.employeename, e.departmentid, e.salary
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee	department	manager
alice smith	engineering	NULL
bob jones	engineering	alice smith
charlie brown	engineering	alice smith
david miller	engineering	bob jones
eva green	engineering	bob jones

Result 4

b. Find employees earning more than their department's average salary. **(2 marks)**

```
56      -- b. Find employees earning more than their department's average salary. (2 marks)
57
58      select e.employeeid, e.employeename, e.departmentid, e.salary
59      from employees e
60      where e.salary > (
61          select avg(sub.salary)
62          from employees sub
63          where sub.departmentid = e.departmentid
64      );
65
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employeeid	employeename	departmentid	salary
1	alice smith	20	150000.00
2	bob jones	20	120000.00
3	charlie brown	20	110000.00
7	grace harris	30	140000.00
10	jack walker	10	95000.00
12	leo allen	40	115000.00
13	mia young	40	85000.00

c. Find the second-highest distinct salary without using LIMIT, TOP, or FETCH FIRST. **(2 marks)**

```
65
66      -- c. Find the second-highest distinct salary without using LIMIT, TOP, or FETCH FIRST. (2 marks)
67
68      select max(salary) as secondhighestsalary
69      from employees
70      where salary < (select max(salary) from employees);
71
72      -- d. Rank employees by salary within each department and display the top three employees per depa
73
74      with rankedemployees as (
75          select
76              employeeid,
77              employeename,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

secondhighestsalary
140000.00

d. Rank employees by salary within each department and display the top three employees per department, retaining ties at the cutoff. **(2 marks)**

```sql
where salary < (select max(salary) from employees);

-- d. Rank employees by salary within each department and display the top three employees per de

with rankedemployees as (
    select
        employeeid,
        employeename,
        departmentid,
        salary,
        dense_rank() over (partition by departmentid order by salary desc) as salaryrank
    from employees
)
select employeeid, employeename, departmentid, salary, salaryrank
from rankedemployees
where salaryrank <= 3;


-- Question 2: Sales Analytics and Query Optimization (6 Marks)
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employeeid	employeename	departmentid	salary	salaryrank
10	jack walker	10	95000.00	1
11	karen hall	10	70000.00	2
1	alice smith	20	150000.00	1
2	bob jones	20	120000.00	2
3	charlie brown	20	110000.00	3
7	grace harris	30	140000.00	1
8	henry clark	30	105000.00	2

Result 7

Question 2: Sales Analytics and Query Optimization (6 Marks)

Create a Sales table with the following columns:

- SaleID

- CustomerID
- SaleDate
- ProductName
- Quantity
- UnitPrice

Insert at least 15 sample records covering multiple customers, products, and dates.

Write SQL queries to:

```sql
-- Question 2: Sales Analytics and Query Optimization (6 Marks)

create table sales (
    saleid int primary key,
    customerid int,
    saledate date,
    productname varchar(100),
    quantity int,
    unitprice decimal(10,2)
);
```

a. Calculate total revenue for each product. **(2 marks)**

```sql
-- a. Calculate total revenue for each product. (2 marks)
select
    productname,
    sum(quantity * unitprice) as total_revenue
from sales
group by productname;
```

b. Calculate monthly revenue and month-over-month revenue change using a window function. **(2 marks)**

```sql
-- b. Calculate monthly revenue and month-over-month revenue change using a window function. (2 marks)
select
    sale_year,
    sale_month,
    monthly_revenue,
    monthly_revenue - lag(monthly_revenue) over (
        order by sale_year, sale_month
    ) as revenue_change
from monthly_sales
order by sale_year, sale_month;
```

c. Rewrite the following query to improve the potential for index usage and explain why your version may perform better. **(2 marks)**

SELECT *

FROM Employees

WHERE YEAR(JoiningDate) = 2025; Assume

JoiningDate may be indexed.

Ans :

SELECT *
FROM Employees
WHERE JoiningDate >= '2025-01-01' AND JoiningDate <= '2025-12-31';

Question 3: SQL Troubleshooting (6 Marks)

A company maintains millions of employee and transaction records. Its reporting queries are running slowly. Perform the following tasks:

a. Explain how an execution plan helps identify expensive operations in a query. **(2 marks) :**

Exposes Data Access Methods: It reveals if the database is using a costly **Full Table Scan** (scanning millions of rows) instead of an efficient **Index Seek/Scan**.

Highlights Costly Operations: It attaches a weight or percentage cost to each individual step (e.g., joins, sorts, aggregations), allowing developers to pinpoint the exact operation consuming the most CPU or I/O.

Flags Mismatches & Bottlenecks: It shows actual versus estimated row counts. Large discrepancies indicate stale statistics, which lead to suboptimal join choices (like a nested loop instead of a hash match).

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b. Recommend an appropriate index strategy for queries filtering employees by DepartmentID and Salary. Explain your reasoning. **(2 marks)** :

CREATE INDEX idx_emp_dept_salary ON Employees (DepartmentID, Salary);

c. Write a query to identify duplicate customer email addresses and display only addresses occurring more than once. **(2 marks)**

```
SELECT EmailAddress, COUNT(EmailAddress) AS OccurrenceCount
FROM Customers
GROUP BY EmailAddress
HAVING COUNT(EmailAddress) > 1;
```

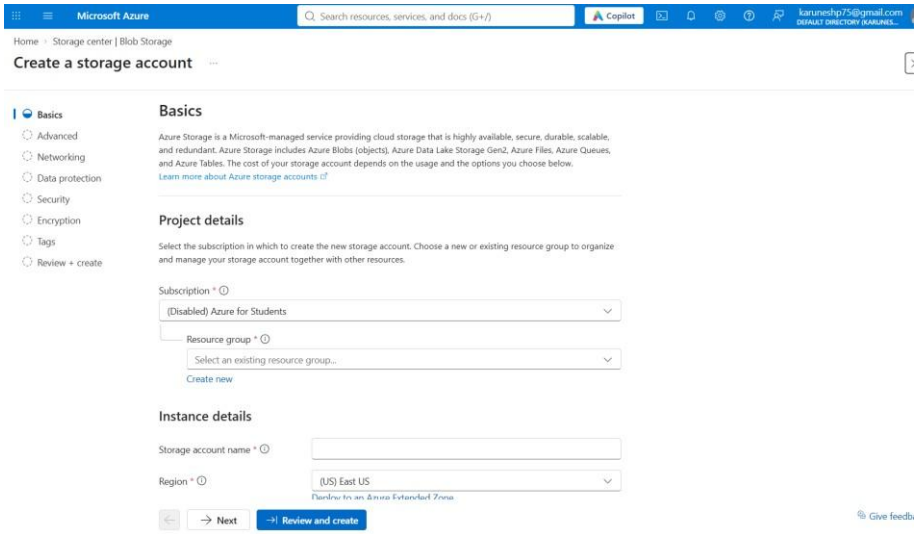
SECTION F: MICROSOFT AZURE || Maximum Marks: 15

Question 15: Azure Data Lake Storage and Security (5 Marks)

A company needs to store confidential business data in Azure for analytics.

Perform or document the following:

a. Configure or describe a storage account with hierarchical namespace enabled for Azure Data Lake Storage Gen2. **(1 mark)**



b. Create a filesystem or container and upload a sample dataset. **(1 mark)**

c. Explain how Microsoft Entra ID and Azure role-based access control can manage access to the data. **(2 marks)** :

Identity Authentication via Entra ID: All users, security groups, or application service principals must authenticate their identity securely using Microsoft Entra ID tokens instead of shared access keys.

Control-Plane Management (Azure RBAC): Azure RBAC roles are applied at the Storage Account or Container level to govern administrative operations. Standard built-in roles include:

Storage Blob Data Owner: Full control over storage container configurations and data.

Storage Blob Data Contributor: Permits reading, writing, and deleting blob/file assets.

Storage Blob Data Reader: Restricts access to read-only capabilities.

d. Explain how you would verify that unauthorized users cannot access the dataset. **(1 mark)** :

1. **Explicit Identity Testing:** Have an unprivileged Entra ID test user or external client attempt to log into the storage account via Azure Storage Explorer, Azure Portal, or a PySpark connection string. The system must immediately throw a 403 Authorization Failed or StorageException: Access Denied error block.
2. **Public Access Verification:** Generate the direct URL endpoint of the uploaded sample file (e.g., https://<account>.dfs.core.windows.net/container/transactions.csv) and attempt to access it through an anonymous browser incognito window. It should return a standard XML/JSON authentication block failure notice, proving anonymous public entry is disabled.
3. **Monitor via Azure Monitor logs:** Inspect **Storage Insights Diagnostic Logs** to verify that failed unauthenticated queries are successfully flagged, categorized, and blocked by the Azure storage security firewall.

Question 16: Azure VM Troubleshooting (5 Marks)

A virtual machine is running in Azure, but administrators cannot connect to it.

Perform or document the following:

a. Verify the VM's provisioning and power states. **(1 mark)**

Provisioning State: This represents the status of the Azure fabric deployment. It should say **Succeeded**. If it says *Failed* or *Updating*, the underlying hardware allocation or extension deployment failed, requiring a redeployment or reapply operation.

Power State: This represents the actual operational runtime state of the hardware. It must display **Running**. If it shows *Stopped* or *Deallocated*, the VM is turned off and cannot accept network traffic.

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b. Explain how Network Security Group rules can prevent connectivity. **(1 mark)**

A **Network Security Group (NSG)** acts as a stateful firewall controlling inbound and outbound traffic. It can block connection attempts through the following mechanics:

Missing Inbound Allow Rules: By default, all inbound traffic from the public internet is blocked by the default **DenyAllInbound** rule. If an explicit rule allowing your connection protocol—such as **Port 22** for Linux SSH or **Port 3389** for Windows RDP—is missing, the connection will drop.

Rule Priority Overrides: NSG rules are evaluated sequentially based on priority numbers (100 to 65000; lower numbers execute first). Even if you have a rule allowing port access at priority 500, a higher-priority rule (e.g., priority 200) that explicitly denies that same IP subnet or port will take precedence and block the traffic.

c. Explain how subnet and routing configuration affect connectivity. **(1 mark)**

Traffic must find a valid path between your client machine and the VM network interface. Connectivity fails if errors exist within the Virtual Network (VNet) routing fabric:

Route Tables & UDRs: Azure automatically provisions system routes to direct traffic. However, if a **User-Defined Route (UDR)** is attached to the VM's subnet that overrides default paths for instance, routing all internet-bound

traffic through a misconfigured or down network virtual appliance (NVA) or firewall the return traffic package gets lost in a routing black hole.

Missing Gateway/Peering Links: If you are trying to access the VM via a private network layout (such as a VPN or ExpressRoute connection), connectivity will drop if **VNet Peering** configurations are broken, or if the subnet's route table lacks a route directing private ranges to the Virtual Network Gateway.

d. Recommend a secure troubleshooting method that avoids unnecessarily exposing the VM to the public internet. **(2 marks)**

Use **Azure Bastion** to establish a secure RDP/SSH session directly inside the Azure Portal browser via TLS/443 without assigning a public IP address to the VM. Alternatively, connect through the native **Serial Console** to troubleshoot lower-level guest OS boot errors over the secure Azure management plane.

Question 17: Azure Data Integration (5 Marks)

A company receives a CSV file containing daily customer transactions and needs to prepare it for analysis.

Design a solution using Azure services.

a. Select an appropriate service to orchestrate data movement. **(1 mark)**

Selected Orchestration Service: Use **Azure Data Factory (ADF)** as the cloud architect to orchestrate data movements. It offers enterprise-grade scheduling, automated workflows, and native serverless connectivity across diverse storage environments.

b. Explain how to copy the CSV file into Azure Data Lake Storage Gen2. **(1 mark)**

Copying to ADLS Gen2: Implement an ADF **Copy Activity** that maps data from your incoming source directory to the destination container. It manages compression, file parsing, and secure, high-throughput cloud loading automatically.

c. Describe how to organize files by ingestion date. **(1 mark)**

I**ngestion Date Organization:** Apply dynamic time expressions within the ADF Sink folder path configurations to generate dated folders on the fly. This automatically segments files into structured paths like year=2026/month=10/day=28/ as they land

d. Explain how to handle malformed records and missing values. **(1 mark)**

Handling Faulty Records: Route data through an ADF **Mapping Data Flow** to clean and process fields before saving. Use *Conditional Split* to isolate malformed rows into a separate error directory and *Derived Column* to substitute missing values with defaults.

e. Explain how to monitor pipeline execution and validate data completeness. **(1 mark)**

Monitoring & Validation: Track runtime pipeline success, pipeline delays, and errors using the visual **ADF Monitor Dashboard** and Azure Monitor logs. Use *Get Metadata* activities to cross-verify row counts and file sizes between source and destination layers.
